On the Guidance of Flow Matching: review on synthetic data

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Abstract

Flow matching learns a deterministic mapping from a simple source distribution to a complex target distribution. It has shown strong performance in image generation and scientific applications such as modeling single-cell perturbation responses. After training, it is often desirable to generate only a specific part of the target distribution without retraining the model. Such a subdistribution can be defined by any user-specified energy function, representing anything from a simple class label to an arbitrarily complex criterion. Guidance methods achieve this by modifying the trained velocity field at inference time. Although recent work has already compared several such methods systematically, the conditions under which their shared assumption holds remain poorly understood. We replicate this comparison of four training-free guidance methods and one learned guidance method on a two-dimensional synthetic benchmark. We evaluate them using distributional metrics that separate fidelity, diversity, and off-manifold errors, with a class-conditional model as reference. Monte Carlo guidance and the learned method perform best and stay competitive with the class-conditional reference. All methods rely on an assumption that only holds exactly when source and target samples are paired independently during training. For Monte Carlo guidance on our benchmark, we show that quality degrades as this pairing gets tighter and the desired subdistribution becomes more selective. One possible application of this analysis, which we do not evaluate here, is guiding flow-matching-based single-cell models toward a rare cell phenotype, since such models are commonly trained with tighter, optimal-transport couplings. Code to reproduce the experiments can be found at https://github.com/scherblu/Guided-Flow-Matching-for-CellOT.

1 Introduction

Deep generative models have achieved remarkable success across domains such as image synthesis (Rombach et al., 2022), computational biology (Klein et al., 2025), and drug discovery (Jumper et al., 2021). A shared principle behind these successes is learning a mapping from a simple, tractable source distribution (e.g. a standard Gaussian) to the complex target distribution of interest. Diffusion models (Ho et al., 2020; Song et al., 2021; Rombach et al., 2022) realize this by learning to reverse a gradual noise-corruption process. Flow matching (Lipman et al., 2023) approaches the same goal more directly. A time-dependent vector field is defined to continuously deform the prior distribution into the target data distribution, and a neural network is trained to approximate this field from data. At inference, new samples are generated by numerically integrating this learned vector field along an ordinary differential equation (ODE) from the source distribution to the target. A critical design choice is the coupling, that is, how individual source samples from the prior are paired with target samples from the data distribution when constructing the training trajectories. Randomly coupling source and target points produces crossing trajectories that place conflicting demands on the vector field and make learning harder, whereas pairing via optimal transport yields shorter, straighter paths

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that are easier to approximate and require fewer ODE integration steps at inference (Tong et al., 2024). Entropic variants of this coupling have further extended flow matching to unpaired data settings such as single-cell genomics (Klein et al., 2024), broadening its practical applicability.

In many applications it is desirable to generate samples that satisfy a user-specified criterion rather than any sample from the target distribution, for instance images of a particular class. One natural approach conditions the model on this criterion at training time, for example on a class label or other side information (Ho and Salimans, 2022). However, this conditioning must be built into the model from the start, and obtaining samples under a new objective requires retraining.

A more flexible alternative is inference-time steering, commonly referred to as guidance. There, the generation process of an already trained model is modified at inference time to bias sampling toward the desired region of the target distribution, without any retraining (Dhariwal and Nichol, 2021; Feng et al., 2025). For diffusion models, this problem is well understood. Classifier guidance (Dhariwal and Nichol, 2021) and classifier-free guidance (Ho and Salimans, 2022) modify the score function at inference time to bias generation toward a desired condition. Transferring these ideas to flow matching is, however, non-trivial, because the deterministic trajectories and the absence of a stochastic noise schedule fundamentally change how an energy function can be incorporated into the velocity field. As a consequence, guidance for flow matching has remained largely underexplored compared to its diffusion counterpart. Feng et al. (2025) recently proposed the first general framework for guidance in flow matching, deriving a family of training-free and training-based techniques covering approximate and asymptotically exact methods. However, a systematic empirical comparison of these approaches across different coupling strengths and a characterization of their failure modes has not been carried out.

We summarize our contributions as follows:

- We provide a review of the guidance methods for flow matching.
- We experimentally test the guidance methods on synthetic data, visualize their behavior, and compare their performance in terms of different metrics.
- We experimentally examine the limitations of the guidance methods regarding different coupling strengths and desired subdistributions and their effect on the generated samples.

2 Background

2.1 Conditional Flow Matching (CFM)

Flow Matching (FM) is a generative method that learns a time-dependent velocity field $v_t(x)$ (with $t \in [0,1]$ and $x \in \mathbb{R}^d$) that transports a tractable source distribution p_0 (e.g. a standard Gaussian) into a complex target distribution p_1 on $\mathbb{R}^d$ (Lipman et al., 2023). The velocity field $v_t(x)$ is a map with $(t,x) \in [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$ since the source and target distributions live in the same space. The velocity field defines a trajectory x_t starting from $x_0 \sim p_0$ through the ordinary differential equation (ODE)

$$\frac{d}{dt}x_t = v_t(x_t), \qquad x_t\big|_{t=0} = x_0. \tag{1}$$

Following this ODE over time from $t=0$ to $t=1$ continuously deforms the source into the target distribution. The distribution of x_t (denoted p_t) coincides with p_0 at $t=0$ and with p_1 at $t=1$. Thus $x_t\big|_{t=1} = x_1 \sim p_1$. A new sample is generated at inference time by drawing $x_0 \sim p_0$ and numerically integrating Eq. (1) forward to $t=1$.

Ideally v_t would be trained by regressing it onto the true marginal vector field $u_t(x)$ that generates the path p_t. This target has no closed form in general. Lipman et al. (2023) sidestep this by conditioning on a latent variable z such as an endpoint pair $z=(x_0,x_1)$ with $x_0 \sim p_0$ and $x_1 \sim p_1$. A conditional probability path $p_t(x|z)$ and a conditional velocity field $u_t(x|z)$ are then chosen in closed form. The marginal path is recovered as

$$p_t(x) = \int p_t(x|z)\, p(z)\, dz \tag{2}$$

and the marginal vector field is the conditional expectation of $u_t(x|z)$ over the posterior $p(z|x_t=x)$ which we can simplify with Bayes' rule to

$$u_t(x) = \int u_t(x|z)\, \frac{p_t(x|z)\, p(z)}{p_t(x)}\, dz. \tag{3}$$

Training v_t to match this marginal target directly amounts to minimizing

$$\mathcal{L}_{FM} = \mathbb{E}_{t\sim\mathcal{U}(0,1),\, x_t\sim p_t}\left[\|v_t(x_t) - u_t(x_t)\|^2\right]. \tag{4}$$

This expectation is intractable in practice. Evaluating $u_t(x_t)$ requires the integral in Eq. (3). Sampling $x_t \sim p_t$ also requires already knowing the path we are trying to learn. They show that regressing v_t onto the closed-form conditional target $u_t(x|z)$ yields the same gradient with respect to v_t as regressing onto the intractable marginal target in Eq. (4). The marginal objective can therefore be replaced by a tractable conditional counterpart. This yields the simulation-free Conditional Flow Matching (CFM) loss

$$\mathcal{L}_{CFM} = \mathbb{E}_{t\sim\mathcal{U}(0,1),\, z\sim p(z),\, x_t\sim p_t(\cdot|z)}\left[\|v_t(x_t) - u_t(x_t|z)\|^2\right], \tag{5}$$

where $u_t(x_t|z)$ is the conditional velocity field that transports x_0 to x_1 along the path defined by $z = (x_0, x_1)$, and $v_t(x_t)$ is the network being trained to imitate the marginal vector field $u_t(x)$. Every variable in Eq. (5) is obtained without ever simulating the ODE in Eq. (1). The time t is drawn uniformly from $[0, 1]$, and the pair $z = (x_0, x_1)$ is sampled from p_0 and p_1 directly or jointly from a coupling as discussed in Section 2.2. Both x_t and $u_t(x_t|z)$ follow directly from the closed-form conditional path given t and z, and $v_t(x_t)$ is simply a forward pass of the network being trained. Training therefore reduces to a plain regression problem.

There are many valid choices for the conditional path $p_t(x|z)$ and vector field $u_t(x|z)$ in Eq. (5). Throughout this paper we assume straight probability paths for all three coupling strategies discussed in Section 2.2. Each trajectory connects x_0 and x_1 along a straight line traversed at constant velocity

$$x_t = (1-t)\,x_0 + t\,x_1, \qquad u_t(x_t|z) = x_1 - x_0 = \frac{x_1 - x_t}{1-t}, \quad t \in [0, 1). \tag{6}$$

We work in the Euclidean space $\mathbb{R}^d$ where a straight line is the shortest connection between two points. Eq. (6) is therefore a reasonable default and is commonly referred to as the "optimal transport" (OT) path. The second equivalent form of u_t in Eq. (6) reappears directly in the velocity corrections used for guidance in Section 3. This choice is not forced by the framework itself. If the data instead lived on a curved manifold, a straight line in the ambient Euclidean space could leave it. Riemannian Flow Matching then replaces Eq. (6) with geodesics (the shortest paths on such spaces) (Chen and Lipman, 2024). Some applications instead call for a stochastic rather than a deterministic generative process. The conditional path can then be built from a Schrödinger bridge instead. This keeps an entropy-regularized transport structure closely related to the entropic coupling introduced in Section 2.2 while allowing genuine randomness along the trajectory (Tong et al., 2024).

2.2 Coupling

Equation (6) only fixes the shape of the path once the pair z is given. It says nothing about how x_0 and x_1 are paired up in the first place. That second choice is called the *coupling* π. A coupling π is any joint distribution over (x_0, x_1) with marginals p_0 and p_1 from which $z \sim \pi$ is drawn instead of the independent sampling $x_0 \sim p_0$ and $x_1 \sim p_1$ used above (Tong et al., 2024). Only the joint distribution π that produces the pair z changes between the three coupling strategies discussed below. The same linear path from Eq. (6) is used throughout. Figure 1 compares the resulting trajectories.

2.2.1 Optimal Transport (OT) Coupling

The most natural choice follows directly from the motivation above. The OT coupling is the coupling π with marginal distributions p_0 and p_1 that minimizes the expected squared transport cost between x_0 and x_1

$$\pi^\star = \arg\min_{\pi}\ \mathbb{E}_{(x_0,x_1)\sim\pi}\left[\|x_1 - x_0\|^2\right]. \tag{7}$$

We use the *minibatch OT* coupling of Tong et al. (2024): for each training batch of n source and n target points, Eq. (7) is solved exactly on just these $2n$ points with the network simplex algorithm, giving the cheapest one-to-one pairing within the batch. Sampling $z \sim \pi^\star$ then pairs each source point with the target point it can reach most cheaply, so trajectories stay short and rarely cross, as seen in the OT-CFM panel of Fig. 1. Such straight non-crossing paths are easier for v_t to approximate and require fewer ODE integration steps at inference (Tong et al., 2024).

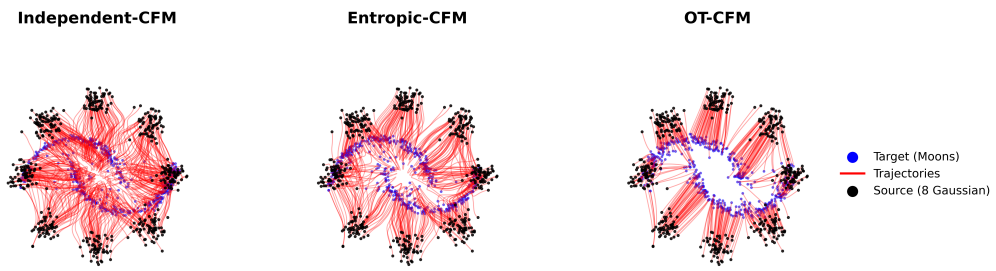


Figure 1: Comparison of three coupling strategies for flow matching: Independent-CFM, Entropic-CFM, and OT-CFM. Each panel shows sample trajectories (red) from the source distribution (8 Gaussians, black) to the target distribution (Moons, blue). OT-CFM produces shorter, straighter trajectories by minimizing transport cost, while Independent-CFM yields crossing paths.

2.2.2 Independent Coupling

The simplest and cheapest alternative avoids solving Eq. (7) altogether by taking the independent coupling $\pi(x_0, x_1) = p_0(x_0)\, p_1(x_1)$ (Tong et al., 2024). This pairs source and target points at random (Independent-CFM panel of Fig. 1). Because the pairing ignores distance, a source point can end up paired with a far-away unrelated target point. Trajectories can then cross one another and place conflicting demands on v_t at the same (t, x). Since v_t must assign a single velocity to every point (t, x), it cannot follow two crossing trajectories at once and instead ends up learning something close to their mean direction, distorting both. This makes the vector field harder to learn and typically requires more ODE integration steps at inference.

2.2.3 Entropic Coupling

Exact OT couplings are costly to recompute for every minibatch and can be unstable to estimate from a finite sample of an empirical distribution. The *entropic coupling* trades some of the straightness of Eq. (7) for robustness and speed by relaxing the transport problem with an added entropy term on π, weighted by a regularization strength $\lambda \in (0, \infty)$. This regularized problem is solved efficiently with the Sinkhorn algorithm rather than an exact assignment solver. As $\lambda \to 0$ the entropic coupling recovers the exact OT coupling, and as $\lambda \to \infty$ it approaches the independent coupling. For instance, we will use $\lambda \in \{0.1, 1.0, 10.0\}$ in our experiments. The intermediate trajectories of the Entropic-CFM panel of Fig. 1 reflect this. This construction extends naturally to unpaired settings such as single-cell genomics (Klein et al., 2024).

The coupling overall controls how straight and non-crossing the training trajectories are. Section 4.4 shows that this choice directly affects how well an already trained flow can be steered by inference-time guidance.

2.3 Class-conditional Flow Matching (CCFM)

The guidance methods introduced in Section 3 steer a flow matching model $v_t(x)$ at inference time, without the model ever seeing the desired subdistribution during training. As a training-time alternative, we consider Class-conditional Flow Matching (CCFM), which instead conditions the velocity field on a discrete label, analogous to conditional generation in diffusion models (Ho and Salimans, 2022) and to class-conditional flow-based generative models trained on labeled image data (Ma et al., 2024). Each target sample x_1 is assigned a label $y = c(x_1) \in \{1, \dots, K\}$ by a fixed partition c of the target distribution. Crucially, y requires no additional sampling step. It is a deterministic function of x_1, which is already part of the pair $z = (x_0, x_1)$ drawn for the ordinary CFM loss, so y is fixed as soon as z is, before x_t is drawn from the conditional path $p_t(\cdot|z)$. The conditional velocity field $v_t(x, y)$ is trained by minimizing

$$\mathcal{L}_{CCFM} = \mathbb{E}_{t\sim\mathcal{U}(0,1),\, z=(x_0,x_1)\sim p(z),\, x_t\sim p_t(\cdot|z)} \left[\| v_t(x_t, c(x_1)) - u_t(x_t|z) \|^2 \right], \tag{8}$$

which is identical to Eq. (5) except that the network additionally receives the label $c(x_1)$ as input. The coupling and path construction of Section 2.2 are unaffected, but the label must be carried through the reshuffling induced by couplings so that $c(x_1)$ stays attached to its source-target pair z.

At inference, a sample from a desired subdistribution is generated by drawing y from the label distribution of that subdistribution and integrating Eq. (1) with $v_t(\cdot, y)$ in place of $v_t(\cdot)$, i.e., no test-time modification of the vector field is required. This makes CCFM substantially less general than the guidance methods of Section 3. It requires the label partition c to be fixed in advance and a separate model to be retrained whenever the target subdistribution changes, and it can only reach subdistributions expressible as a mixture of the fixed classes. In exchange, because the conditioning is learned rather than approximated at inference, we use CCFM in Section 4 as a class-conditional baseline against which the training-free guidance methods are compared.

3 Guidance

This section presents the guidance methods for flow matching introduced by Feng et al. (2025). Guidance modifies an already trained flow matching model at inference time so that it generates samples from a desired subdistribution of the target distribution p_1 without retraining the underlying model. Following Feng et al. (2025), we look for a guidance vector field $g_t(x_t)$ that is added to the trained velocity field, $v'_t(x_t) = v_t(x_t) + g_t(x_t)$, so that the modified field transports p_0 to the desired subdistribution instead of p_1. Which subdistribution is desired is expressed through a scalar energy function J, mapping $\mathbb{R}^d$ to $\mathbb{R}$, that assigns low values to preferred outcomes and high values to everything else, so that only the low-energy region is effectively generated. A concrete example of such an energy function, together with the subdistribution of p_1 it selects in our experiments, is given in Section 4. Low-energy regions are exponentially up-weighted while the rest of p_1 is left untouched, so J can encode anything from selecting a subset of the modes of p_1 to steering generation toward a desired property more generally. Formally, this amounts to sampling from a tilted target distribution

$$p'(x) = \frac{1}{Z}\, p_1(x)\, e^{-J(x)}, \qquad Z = \int p_1(x)\, e^{-J(x)}\, dx, \tag{9}$$

where Z is simply the normalizing constant that makes p' a valid probability distribution. Producing samples from p' instead of p_1 also means deciding how each new x_1 is paired with a source point x_0. Analogously to p' above, reweighting the joint coupling $\pi(x_0, x_1)$ used during training defines a new joint coupling for the guided process,

$$\pi'(x_0, x_1) = \frac{1}{Z}\, \pi(x_0, x_1)\, e^{-J(x_1)}. \tag{10}$$

Feng et al. (2025) compare the reverse coupling implied by π' to the one implied by the training coupling π through the reverse coupling ratio

$$\mathcal{P} = \frac{\pi'(x_0|x_1)}{\pi(x_0|x_1)}. \tag{11}$$

This ratio is exactly one for the independent coupling of Section 2.2.2, since there $\pi(x_0|x_1) = p_0(x_0)$ does not depend on x_1, and only approximately one for the OT and entropic couplings, where the pairing does depend on x_1. Accounting for $\mathcal{P}$, the exact guidance field takes the form

$$\begin{aligned} g_t(x_t) &= \int \left(\mathcal{P}\, \frac{e^{-J(x_1)}}{Z_t(x_t)} - 1 \right) u_t(x_t|z)\, p(z|x_t)\, dz \\ &= \mathbb{E}_{z\sim p(z|x_t)} \left[\left(\mathcal{P}\, \frac{e^{-J(x_1)}}{Z_t(x_t)} - 1 \right) u_t(x_t|z) \right], \\ Z_t(x_t) &= \int \mathcal{P}\, e^{-J(x_1)}\, p(z|x_t)\, dz \\ &= \mathbb{E}_{z\sim p(z|x_t)} \left[\mathcal{P}\, e^{-J(x_1)} \right], \end{aligned} \tag{12}$$

where the integral can be rewritten as an expectation over the posterior $p(z|x_t)$ of the latent variable $z = (x_0, x_1)$ given the current state x_t. Following Feng et al. (2025), we assume $\mathcal{P} = 1$ for the

remainder of this section, which simplifies Eq. (12). This expression still cannot be evaluated directly, even under $\mathcal{P} = 1$. The posterior $p(z|x_t)$, the distribution over endpoints $z = (x_0, x_1)$ that are consistent with the current state x_t, has no closed form. The four methods below differ only in how they approximate this posterior or the resulting expectation, trading accuracy for computational cost, and we return to the effect of the $\mathcal{P} = 1$ assumption itself in Section 4.4.

3.1 Monte Carlo (MC) estimation

MC guidance targets the exact guidance field of Eq. (12) directly. By Bayes' rule, the intractable posterior can be written as $p(z|x_t) = p_t(x_t|z)\, p(z)/p_t(x_t)$, where $p(z) = \pi(x_0, x_1)$ is precisely the coupling used to train the model, be it the OT, entropic, or independent coupling of Section 2.2, and is therefore easy to sample from regardless of which one was used. Substituting this into Eq. (12) turns the expectation over the intractable posterior into an expectation over the tractable coupling $p(z)$, reweighted by $p_t(x_t|z)/p_t(x_t)$. Both this reweighted expectation and the normalizing constant $Z_t(x_t)$ are then approximated by Monte Carlo estimation, drawing N pairs $z^i = (x_0^i, x_1^i) \sim p(z)$ once and reusing them to self-normalize $p_t(x_t)$, which gives the estimator

$$g_t^{MC}(x_t) = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{e^{-J(x_1^i)}}{\hat{Z}_t(x_t)} - 1\right) u_t(x_t|z^i)\,\frac{p_t(x_t|z^i)}{\hat{p}_t(x_t)}, \qquad \hat{p}_t(x_t) = \frac{1}{N}\sum_{i=1}^{N} p_t(x_t|z^i), \quad (13)$$

with $\hat{Z}_t(x_t)$ estimated the same way from the same N samples. As $N \to \infty$, this estimator converges to the exact field of Eq. (12), so g^{MC} makes no assumption on the source distribution or on the coupling. Its price is variance. Once J is even moderately selective, most sampled x_1^i receive a near-zero weight $e^{-J(x_1^i)}$, so a large number of samples is needed for a stable estimate, and this requirement grows quickly with the dimensionality of the data. Hence, this MC method is not computationally efficient in practice for high-dimensional data and selective energy functions.

3.2 Simulation-based Monte Carlo estimation

Rather than sampling from the training coupling and hoping enough samples land near the low-energy region of J, simulated (sim-)MC guidance approximates the posterior itself with a Gaussian $\tilde{p}(x_1|x_t) = \mathcal{N}(x_1;\, \hat{x}_1(x_t), \Sigma_t(x_t))$ fitted to match its first two moments, and draws samples directly from it,

$$g_t^{\text{sim-MC}}(x_t) = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{e^{-J(x_1^i)}}{\tilde{Z}_t} - 1\right) u_t(x_t|z^i), \qquad x_1^i \sim \tilde{p}(x_1|x_t). \quad (14)$$

Because samples are now drawn already concentrated near the relevant region, far fewer of them are needed than for g^{MC}. This comes at the cost of a bias introduced by the Gaussian approximation, which can be poor when the true posterior $p(x_1|x_t)$ has multiple modes.

3.3 Covariance Guidance (cov-A / cov-G)

A cheaper alternative common in the diffusion literature avoids sampling altogether, at the cost of assuming that the posterior $p(x_1|x_t)$ is concentrated around its mean $\hat{x}_1(x_t) := \mathbb{E}_{z\sim p(z|x_t)}[x_1]$. Feng et al. (2025) estimate $\hat{x}_1(x_t)$ cheaply from the current network prediction under the *affine path assumption*, namely that the conditional path takes the form $x_t = \alpha_t x_1 + \beta_t x_0 + \sigma_t \varepsilon$ for some schedules $\alpha_t, \beta_t, \sigma_t$ and noise ε. This assumption is satisfied exactly by the straight-line path of Eq. (6), with $\alpha_t = t$, $\beta_t = 1 - t$, and $\sigma_t = 0$, which gives $\hat{x}_1(x_t) = x_t + (1 - t)\, v_t(x_t)$. A first-order Taylor expansion of $e^{-J(x_1)}$ around $\hat{x}_1$ then turns the intractable expectation of Eq. (12) into a gradient of J evaluated at this single point estimate, preconditioned by the covariance $\Sigma_{1|t}$ of $p(x_1|x_t)$,

$$g_t^{cov}(x_t) \propto \Sigma_{1|t}\, \nabla_{\hat{x}_1} J(\hat{x}_1). \quad (15)$$

The two variants considered here differ only in how $\Sigma_{1|t}$ is obtained.

- **cov-A** replaces it with a hand-tuned decaying schedule of scaling the identity matrix $\lambda_t I$, so the guidance reduces to a rescaled gradient of J that requires no extra network evaluations.
- **cov-G** instead computes $\Sigma_{1|t}$ exactly under the additional assumption that the source distribution is Gaussian and independent of the target.

3.4 Learned guidance (GM)

The methods above are training-free and are evaluated at sampling time on top of a fixed base model. Learned guidance instead amortizes the cost of guidance into a one-time training step. A network g_ϕ, together with an auxiliary network $Z_\phi \approx Z_t$, is fitted with a regression loss whose minimizer provably coincides with the exact field of Eq. (12),

$$\mathcal{L}_\phi = \mathbb{E}_{t,\, z,\, x_t}\left[\left\| g_\phi(x_t, t) - \left(\tfrac{e^{-J(x_1)}}{Z_\phi(x_t,t)} - 1\right) u_t(x_t|z)\right\|_2^2\right]. \tag{16}$$

This is mirroring how the conditional flow matching loss of Eq. (5) replaces an intractable marginal objective with a tractable conditional one. Once trained, g_ϕ is a single cheap forward pass at inference time. The trade-off is that a new pair of networks must be trained for every energy function J, unlike the training-free methods above, which can be reused across arbitrarily many choices of J on the same base model.

3.5 Discussion

The four methods trade off bias, variance, and computational cost differently. The MC method makes the fewest assumptions and needs no additional training, but its variance grows quickly with the dimensionality of the data and the selectivity of J. Learned guidance removes this variance by shifting the cost to a one-time training step, at the price of retraining whenever J changes. The covariance methods are the cheapest at inference and need no sampling at all, but they rest on the assumption that $p(x_1|x_t)$ is well summarized by a point estimate and a covariance, an assumption that can fail once J carves out a small or oddly shaped region of p_1. We compare all methods empirically in Section 4 and examine in Section 4.4 how their accuracy depends on the strength of the coupling used during training.

4 Results

4.1 Synthetic data and energy function

We evaluate the guidance methods of Section 3 on a two-dimensional synthetic benchmark, which lets us visualize the guided samples directly and compare them against a known ground truth. The target distribution p_1 is a mixture of eight Gaussians arranged on a circle, and the source distribution p_0 is uniform. We use this uniform-to-eight-Gaussian pair as the running example for most of our experiments, since it is easy to visualize and its eight well-separated modes give a clear target for guidance. Unless stated otherwise, the base CFM model is trained on this source-target pair with the OT coupling of Section 2.2.

To define a subdistribution to guide toward, we pick three of the eight modes with centers c_0, c_2, c_3 and build an energy function that is low near any of these three centers and high everywhere else

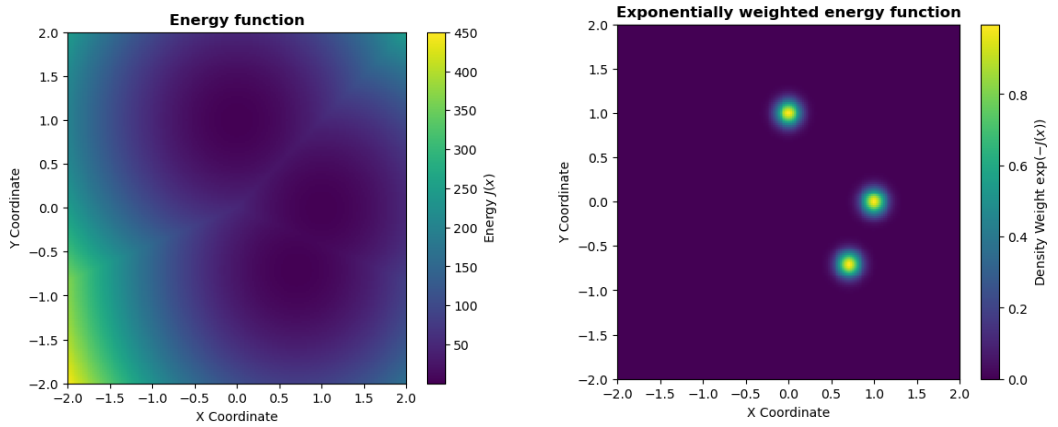
$$J(x) = -\log \sum_{i\in\{0,2,3\}} \exp\left(-\frac{\|x - c_i\|_2^2}{2\sigma^2}\right). \tag{17}$$

The width σ controls how selective J is. A small σ only rewards points very close to a center, while a large σ makes the guided target closer to the full eight-mode mixture. Unless stated otherwise, we set $\sigma = 0.1$, which keeps the three selected modes clearly separated from the rest of the mixture without making J so sharp that guidance becomes extremely selective. Figure 2 shows this energy function together with the resulting weight $e^{-J(x)}$, which is proportional to the guided target density of Eq. (9). The three low-energy areas around the selected centers are clearly separated from each other, so a good guidance method should place its generated samples only inside these areas.

Base model architecture and training details are given in Appendix A.1, and further guidance hyperparameter tuning is discussed in Appendix A.4.

4.2 Qualitative comparison of guidance methods

We first compare the seven methods visually, before turning to quantitative metrics next. Figure 3 shows samples from the unguided base model (CFM), the four training-free guidance methods,



(a) Energy function $J(x)$.

(b) Exponentially weighted energy $e^{-J(x)}$.

Figure 2: Left, the energy function $J(x)$ of Eq. (17) selecting three of the eight Gaussian modes, where low values mark the preferred region. Right, the corresponding weight $e^{-J(x)}$, proportional to the guided target $p'(x) \propto p_1(x)\, e^{-J(x)}$. The three separated peaks show that only samples near the selected modes are preferred once guidance reweights the target distribution.

learned guidance (GM), and the class-conditional flow matching (CCFM) method, each applied to three energy functions. The first row guides a uniform-to-eight-Gaussian flow toward a single mode. The second row guides the same flow toward the three modes defined in Eq. (17). The third row guides an eight-Gaussian-to-moons flow toward a contiguous arc rather than a set of isolated modes.

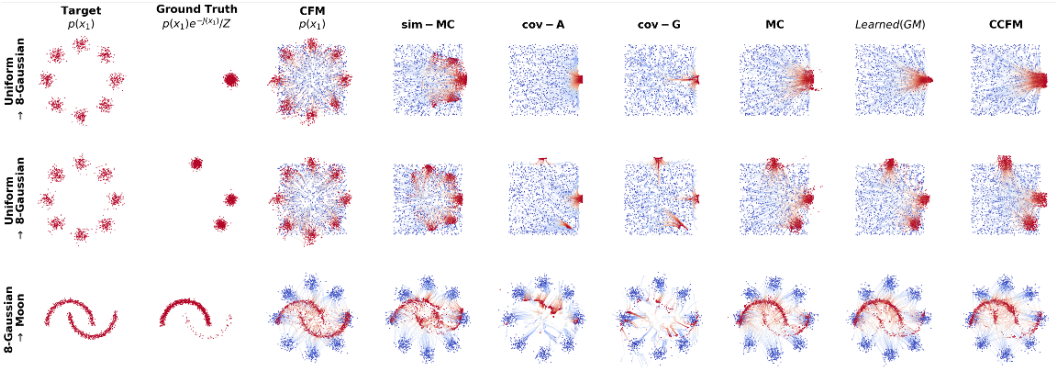


Figure 3: Qualitative comparison of guidance methods across three energy functions (rows), guiding toward one mode, toward three modes, and toward a contiguous arc of the moons distribution. Columns show the target distribution, the ground truth guided density $p'(x)$, and samples generated by the unguided base model and each guidance method. Blue points belong to the source distribution and red points belong to the target distribution. The path between each coupling is colored from blue to red along its way.

Across all three rows, the unguided base model reproduces the full target distribution and ignores the energy function, confirming that the guided methods are doing genuine work rather than the base model already matching the target region by chance. Learned GM concentrates the most samples inside the ground truth region with the fewest visible outliers, closely matching CCFM despite never seeing the labels during training. MC also places most samples correctly, but leaves a visible trail of points along the transport paths that have not fully reached the target region yet. The two covariance methods behave quite differently from each other for the 8gaussian to moons transition. Cov-A collapses onto a narrow line through the selected modes and undershoots the spread of the ground truth, while cov-G spreads more broadly but also leaves more points scattered outside the selected region. Sim-MC shows the most scattered outliers among the guided methods, which is most visible

in the third row, where the arc-shaped target region is harder to approximate with a single Gaussian than a compact cluster.

In short, the qualitative comparison already suggests a rough ranking. GM and MC concentrate samples most reliably inside the guided region, the two covariance methods each trade concentration for a different kind of error, and sim-MC becomes unreliable once the target region stops being a compact blob. We quantify this ranking next.

4.3 Quantitative comparison of guidance methods

We now measure the three-mode setting of Figure 3 with four complementary metrics, each capturing a different way a guidance method can fail. The Wasserstein-2 (W2) distance measures the overall transport cost between generated and ground truth samples (Peyré and Cuturi, 2020). The energy distance measures whether the two point clouds occupy the same region of space (Székely and Rizzo, 2013). The diversity trace ratio compares the spread of the generated samples to the spread of the ground truth, so a value below one means the guided samples are too concentrated and a value above one means they are too spread out. The off-manifold rate measures how often a generated sample lands far from any real data point, that is, how often guidance produces an unrealistic outlier. Formal definitions of all four metrics are given in Appendix A.2.

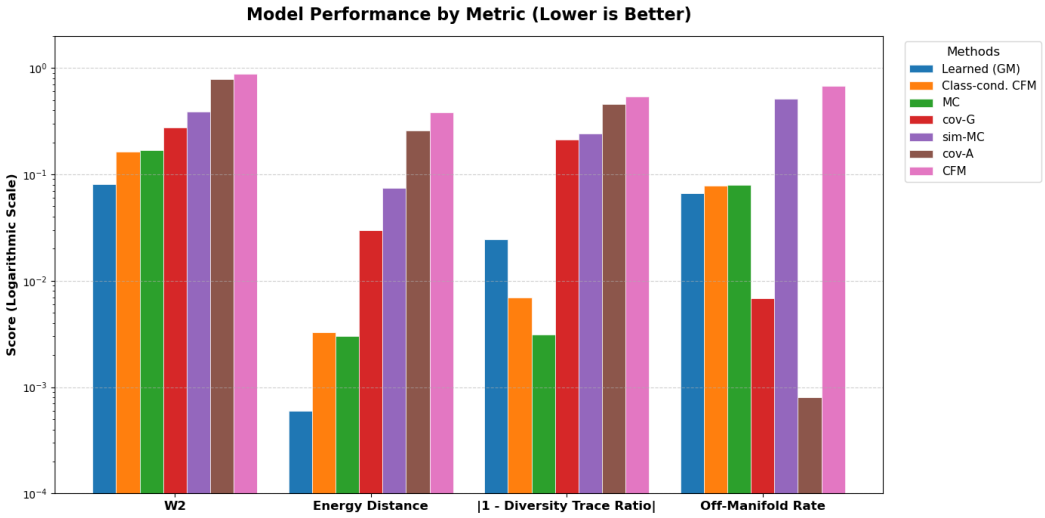


Figure 4: Quantitative comparison of all seven methods on the three-mode setting of Figure 3, using four metrics on a logarithmic scale where lower is always better. Learned guidance (GM) reaches the lowest W2 distance and energy distance. MC guidance offers the best overall balance among the training-free methods.

Figure 4 reports all four metrics on a logarithmic scale, with exact values listed in Table 1. For the W2 distance, energy distance, and off-manifold rate, lower is better. For the diversity trace ratio, we plot the absolute difference from the ideal value of one, so lower is better throughout. GM reaches the best W2 distance (0.080) and energy distance (0.0006), roughly five times lower than both class-conditional flow matching and MC guidance on the energy distance. This is coherent with the qualitative comparison of Figure 3, where learned GM produces the most concentrated samples inside the low-energy region. MC guidance trails GM by roughly the same five-fold margin on the energy distance, but it closely matches class-conditional flow matching instead: within a few percent on W2 (0.170 versus 0.163), within a similar margin on energy distance (0.0030 versus 0.0033), and slightly better on the diversity trace ratio (0.997 versus 1.007), all without ever training on the target labels. The covariance methods each trade accuracy for a low off-manifold rate. Cov-A reaches the lowest off-manifold rate of all seven methods (0.0008), but its diversity trace ratio of 0.541 confirms the mode collapse already visible in Figure 3. Cov-G keeps a moderate off-manifold rate (0.007) but over-disperses instead, with a diversity trace ratio of 1.211, and its W2 distance is multiple times worse than GM or MC. Despite its low off-manifold rate, cov-A is in fact worst among the guided methods on the other three metrics, W2, energy distance, and diversity trace ratio, since its mode

collapse concentrates samples near a few points that still happen to sit close to real data. Sim-MC instead has the worst off-manifold rate of any guided method (0.510, more than five hundred times higher than cov-A), which matches its visibly scattered outliers in Figure 3, even though it trails cov-A specifically on the other three metrics. The unguided base model performs worst overall, as expected since it does not use the energy function at all.

Overall, the learned GM and MC guidance methods are the most accurate, with GM slightly better than MC but requiring retraining for every energy function. The covariance methods perform best in terms of off-manifold rate, i.e. less outliers due to mode collapse but sacrifice the other metrics. The sim-MC's Gaussian approximation of the posterior is the most fragile of the four training-free methods once the target region is no longer a single compact cluster.

4.4 Limitations of guidance

All four guidance methods of Section 3 rely on the assumption $\mathcal{P} = 1$ on the reverse coupling ratio, which holds exactly only for the independent coupling of Section 2.2.2. In practice, base models are often trained with the OT or entropic coupling instead, because the resulting straighter trajectories are cheaper to integrate at inference. We test how much this mismatch matters for MC guidance, the training-free method with the best overall balance in Figure 4. We train separate base models with coupling strength ranging from the independent coupling through entropic regularization at several strengths to the exact OT coupling, and we vary the width σ of the energy function in Eq. (17) from sharp ($\sigma = 0.03$) to broad ($\sigma = 0.2$).

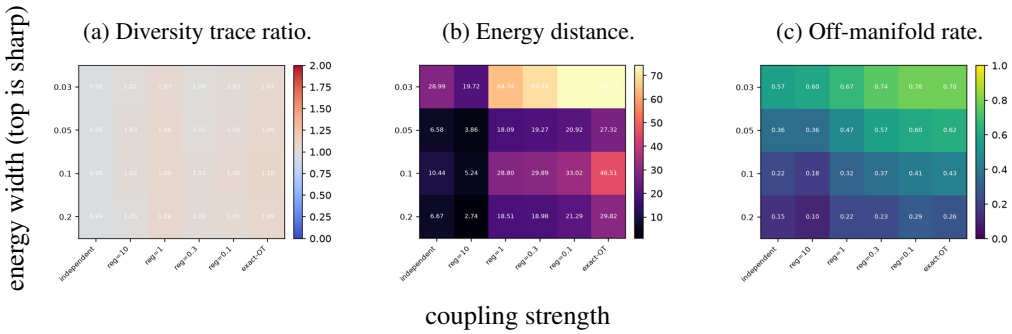


Figure 5: Phase diagrams for MC guidance across coupling strength (columns, from the independent coupling to exact OT) and the width σ of the energy function (rows, from sharp at the top to broad at the bottom). Stronger coupling consistently increases the energy distance and the off-manifold rate, while the diversity trace ratio stays close to one throughout, showing that the independence assumption behind MC guidance degrades gradually rather than suddenly as the training coupling tightens.

Figure 5 shows the three metrics over this grid. The diversity trace ratio stays close to one everywhere, between 0.98 and 1.10, so MC guidance preserves the spread of the guided target regardless of coupling strength or energy sharpness. The energy distance grows substantially as the coupling gets tighter. At every energy width we tested, it rises by roughly a factor of four to five when moving from the independent coupling to exact OT, for instance from 6.58 to 27.32 at $\sigma = 0.05$. The off-manifold rate also grows with coupling strength at every energy width, for example from 0.15 to 0.26 at the broadest setting and from 0.57 to 0.78 at the sharpest, increases of roughly forty to seventy percent. The off-manifold rate additionally depends on how selective the energy function is. Comparing the sharpest row to the broadest row at any fixed coupling strength roughly triples the off-manifold rate.

These three metrics together show that MC guidance degrades gradually as the $\mathcal{P} = 1$ assumption is violated more strongly. The generated samples keep roughly the right spread, but they move further from the true guided target and land off the data manifold more often, and both effects get worse the more selective the desired subdistribution is. Applying training-free guidance on top of a model trained with strong OT or entropic coupling, which is common in practice, therefore needs more caution than applying it on top of an independently coupled model, especially when the desired subdistribution is narrow.

4.5 Limitations of this study

All results above are from a single run without repeated seeds or reported variance, so the close calls between MC guidance and CCFM on the W2 distance and the diversity trace ratio in Section 4 are only suggestive. Confirming which method is actually better would require multiple seeds and a variance estimate. The coupling-strength stress test is similarly narrow in scope, covering only MC guidance on a two-dimensional synthetic benchmark. The finding that guidance degrades as training coupling tightens has therefore not been verified for the other three methods or for higher-dimensional, more realistic settings such as single-cell trajectory models. Feng et al. (2025) themselves report notable performance gaps between their two-dimensional toy experiments and their more complex experiments on image inverse problems and offline reinforcement learning, so such a generalization should not be assumed. We restricted the sweep to MC because it already gave the best training-free balance in Section 4. Cov-A was tried as well but excluded here since it already collapses onto a narrow line regardless of coupling strength (diversity trace ratio 0.541, Table 1), so sweeping it over coupling strength would mostly reflect this pre-existing mode collapse rather than isolate the effect of $\mathcal{P} \neq 1$. We expect cov-G and GM to degrade similarly to MC since all methods share the same $\mathcal{P} = 1$ approximation, though GM's learned regression objective could in principle absorb some of this mismatch during training rather than only expressing it at inference, and this remains untested.

5 Conclusion

We reviewed and empirically compared the guidance methods for flow matching proposed by Feng et al. (2025) on synthetic data. Among the training-free methods, MC guidance offers the best overall balance across distributional metrics, while learned guidance (GM) gives a further, modest improvement at the cost of retraining for every energy function. Both guidance methods remain competitive with class-conditional flow matching (CCFM) despite never observing the target subdistribution during training. However, all four methods rely on the $\mathcal{P} = 1$ assumption, which only holds exactly for the independent coupling, and our phase-diagram analysis of MC guidance shows that quality degrades as the training coupling tightens toward exact OT and as the energy function becomes more selective. Non-independent couplings such as OT are common in practice, so this mismatch deserves caution whenever guidance is applied on top of such a model. One possible application we did not evaluate here is guiding flow-matching-based single-cell models, which are commonly trained with such couplings, toward a rare cell phenotype.

References

Ricky T. Q. Chen and Yaron Lipman. Flow matching on general geometries. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=g7ohDlTITL.

Prafulla Dhariwal and Alexander Quinn Nichol. Diffusion models beat gans on image synthesis. In *Advances in Neural Information Processing Systems*, volume 34, pages 8780–8794, 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/hash/49ad23d1ec9fa4bd8d77d02681df5cfa-Abstract.html.

Ruiqi Feng, Chenglei Yu, Wenhao Deng, Peiyan Hu, and Tailin Wu. On the guidance of flow matching. In Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu, editors, *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 16993–17029. PMLR, 13–19 Jul 2025. URL https://proceedings.mlr.press/v267/feng25s.html.

Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. In *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications*, 2022. URL https://openreview.net/forum?id=qw8AKxfYbI.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851, 2020. URL https://proceedings.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html.

John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, Anna Potapenko, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873):583–589, 2021. doi: 10.1038/s41586-021-03819-2. URL https://doi.org/10.1038/s41586-021-03819-2.

Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.

Dominik Klein, Théo Uscidda, Fabian Theis, and Marco Cuturi. Genot: Entropic (gromov) wasserstein flow matching with applications to single-cell genomics. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 103897–103944. Curran Associates, Inc., 2024. doi: 10.52202/079017-3301. URL https://proceedings.neurips.cc/paper_files/paper/2024/file/bc46e29f91e676747c584ca181cb0ea1-Paper-Conference.pdf.

Dominik Klein, Jonas Simon Fleck, et al. CellFlow enables generative single-cell phenotype modeling with flow matching. *bioRxiv*, 2025. doi: 10.1101/2025.04.11.648220. URL https://www.biorxiv.org/content/10.1101/2025.04.11.648220v1.

Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=PqvMRDCJT9t.

Nanye Ma, Mark Goldstein, Michael S. Albergo, Nicholas M. Boffi, Eric Vanden-Eijnden, and Saining Xie. Sit: Exploring flow and diffusion-based generative models with scalable interpolant transformers. In *European Conference on Computer Vision*, 2024. URL https://arxiv.org/abs/2401.08740.

Gabriel Peyré and Marco Cuturi. Computational optimal transport, 2020. URL https://arxiv.org/abs/1803.00567.

Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695, 2022. URL https://openaccess.thecvf.com/content/CVPR2022/html/Rombach_High-Resolution_Image_Synthesis_With_Latent_Diffusion_Models_CVPR_2022_paper.html.

Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021. URL https://openreview.net/forum?id=PxTIG12RRHS.

Gábor J. Székely and Maria L. Rizzo. Energy statistics: A class of statistics based on distances. *Journal of Statistical Planning and Inference*, 143(8):1249–1272, 2013. ISSN 0378-3758. doi: https://doi.org/10.1016/j.jspi.2013.03.018. URL https://www.sciencedirect.com/science/article/pii/S0378375813000633.

Alexander Tong, Kilian FATRAS, Nikolay Malkin, Guillaume Huguet, Yanlei Zhang, Jarrid Rector-Brooks, Guy Wolf, and Yoshua Bengio. Improving and generalizing flow-based generative models with minibatch optimal transport. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL https://openreview.net/forum?id=CD9Snc73AW. Expert Certification.

A Appendix

A.1 Base model training details

All base flow matching models in this paper, CFM and its class-conditional variant CCFM, share the same architecture and optimization recipe. Only the source, target, and training coupling of Section 2.2 differ across experiments.

Architecture. Following Feng et al. (2025), we parameterize the velocity field with an MLP that has 3 hidden layers of width 256 and SELU activations, mapping the 2-dimensional state x_t concatenated with time t to a 2-dimensional velocity, for 133k parameters in total. CCFM additionally concatenates a learned class embedding to the input.

Optimization. We minimize the CFM loss of Eq. (5) (or Eq. (8) for CCFM) with the deterministic straight-line conditional path of Eq. (6), using Adam (Kingma and Ba, 2015) at a learning rate of 10^{-4} for 60,000 steps. Every step draws a fresh batch of 1024 pairs (x_0, x_1) from the source and target distributions under the chosen coupling, so training uses an effectively unlimited synthetic data stream rather than a fixed-size dataset.

A.2 Definition of evaluation metrics

We evaluate every guidance method by comparing n generated samples $X = \{x_i\}_{i=1}^n$ against m ground truth samples $Y = \{y_j\}_{j=1}^m$ drawn from the guided target p' of Eq. (9). Non-finite samples (e.g., from a diverging ODE solve) are removed from X before computing W2, energy distance, and the diversity ratio, and are instead counted as off-manifold outliers.

Wasserstein-2 (W2) distance. Optimal transport cost between the empirical distributions $\hat{p}_X, \hat{p}_Y$ under squared Euclidean cost, solved with the POT library (Peyré and Cuturi, 2020):

$$W_2(X, Y) = \left(\min_{\gamma \in \Pi(\hat{p}_X, \hat{p}_Y)} \sum_{i=1}^{n} \sum_{j=1}^{m} \gamma_{ij} \, \|x_i - y_j\|_2^2 \right)^{1/2}. \tag{18}$$

Smaller is better; it jointly captures location and spread mismatch between X and Y.

Energy distance. Compares average pairwise distances within and between X and Y (Székely and Rizzo, 2013), estimated as the V-statistic

$$\hat{D}_E(X, Y) = \frac{2}{nm} \sum_{i=1}^{n} \sum_{j=1}^{m} \|x_i - y_j\|_2 - \frac{1}{n^2} \sum_{i=1}^{n} \sum_{i'=1}^{n} \|x_i - x_{i'}\|_2 - \frac{1}{m^2} \sum_{j=1}^{m} \sum_{j'=1}^{m} \|y_j - y_{j'}\|_2, \tag{19}$$

clamped at zero. $\hat{D}_E \geq 0$, with $\hat{D}_E = 0$ iff X and Y occupy the same region of space.

Diversity trace ratio. Ratio of the traces of the empirical covariance matrices of X and Y,

$$R_{\text{div}}(X, Y) = \operatorname{tr}\big(\widehat{\operatorname{Cov}}(X)\big) \,/\, \operatorname{tr}\big(\widehat{\operatorname{Cov}}(Y)\big). \tag{20}$$

A ratio below one indicates mode collapse from guidance; above one indicates over-dispersion.

Off-manifold rate. Fraction of generated samples whose nearest ground-truth neighbor is farther than a threshold τ, the q=0.99 quantile of ground truth's own intra-set nearest-neighbor distances:

$$\text{OMR}(X, Y) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}[d_i > \tau], \qquad d_i = \min_j \|x_i - y_j\|_2. \tag{21}$$

Non-finite samples removed from X beforehand always contribute 1 to this sum, i.e., count as off-manifold.

A.3 Result data in tabular form

Table 1: Comparison of methods across various metrics. The best method for each metric is **bolded**, and the second best is underlined.

Method	W2 ↓	Energy Dist ↓	Div Trace Ratio	Off-Manifold Rate ↓
Learned (GM)	**0.0802**	**0.0006**	0.9757	0.0659
Class-cond. CFM	0.1632	0.0033	1.0069	0.0786
MC	0.1698	0.0030	**0.9969**	0.0801
cov-G	0.2772	0.0296	1.2113	0.0068
sim-MC	0.3908	0.0739	0.7570	0.5098
cov-A	0.7905	0.2578	0.5410	**0.0008**
CFM	0.8852	0.3804	1.5355	0.6782

A.4 Tuning hyperparameters

A.4.1 ODE solver

Increasing the number of steps in the ODE solver from 100 to 200 improved the performance of the guidance methods. A higher-order solver such as Runge-Kutta 4th order could improve samples further at the cost of more computation, and an early stopping criterion could mitigate numerical instabilities at $t = 1$ and mode collapse in MC guidance.

A.4.2 MC Guidance

The parameter `mc_ep` (ϵ) appears in the denominator $(1 - t + \epsilon)$ of both the MC importance weights and the velocity correction $u = (x_1 - x_t)/(1 - t + \epsilon)$, acting as a numerical stabilizer as $t \to 1$. Without sufficient ϵ, back-projected source points fall outside the prior support, causing all importance weights to clamp to a uniform distribution and injecting large random forces that scatter particles off the manifold. We tried $\epsilon \in \{0.001, 0.005, 0.01, 0.05, 0.1\}$, with results shown in Table 2. Hence, we increased `mc_ep` from 0.001 to 0.05, which reduced the numerical instability and the outliers of the generated distribution significantly.

Table 2: Effect of the `mc_ep` parameter ϵ on sample quality.

ϵ	W_2 Distance ↓	Off-Manifold Rate ↓
0.001 (default)	0.4244	0.266
0.010	0.3053	0.042
0.050 (tuned)	**0.1091**	**0.002**
0.100	0.2032	0.004

A.4.3 Number of samples for computing metrics

We report all metrics in Figure 4 and Table 1 using 2048 generated points compared against 2048 ground truth points per method. Feng et al. (2025) show that the W2 distance converges by 500 samples, so 2048 points are sufficient for stable metrics.